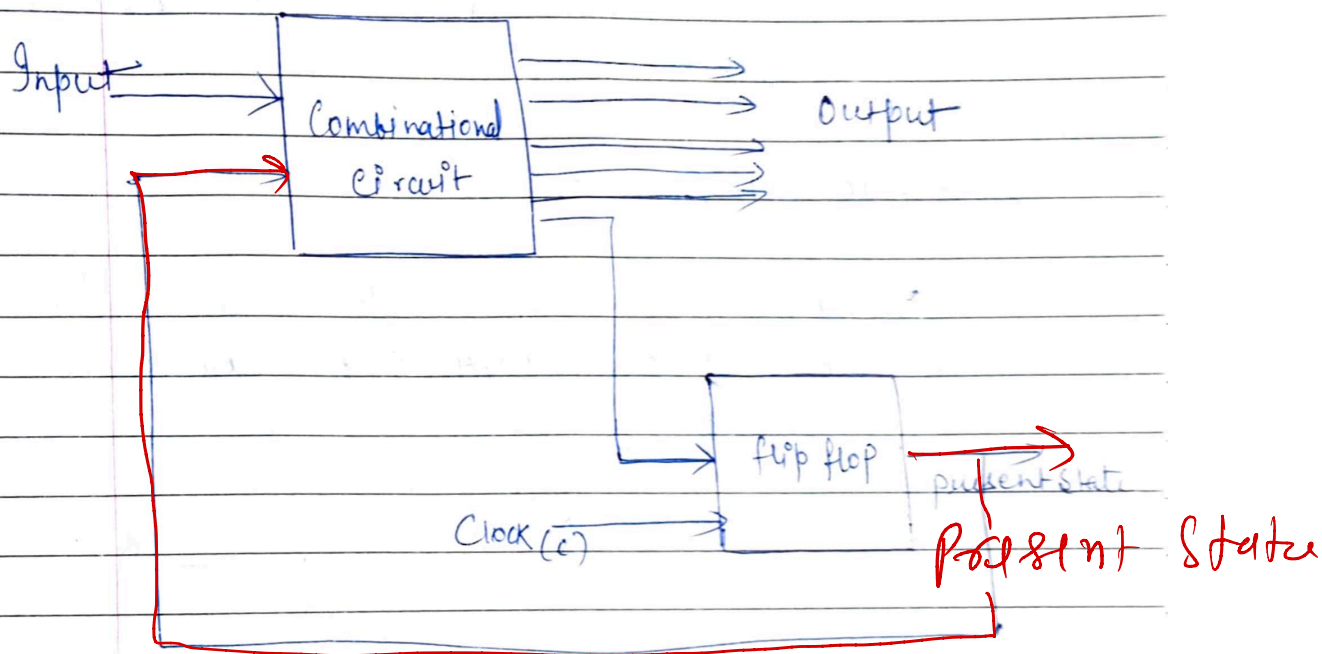


SEQUENTIAL CIRCUIT:-

It is an interconnection of combinational circuits & storage elements (flip flop) that stores binary information that indicates the state of sequential circuit at the time.

Block diagram of Sequential circuit



Sequential circuits are time dependent. It is of two types i.e. categorised on the basis the time at which these are observed and their internal state changes.

- (a) Synchronous
- (b) Asynchronous

(a) Synchronous → These circuit use flip flop and their status can change only at discrete intervals.

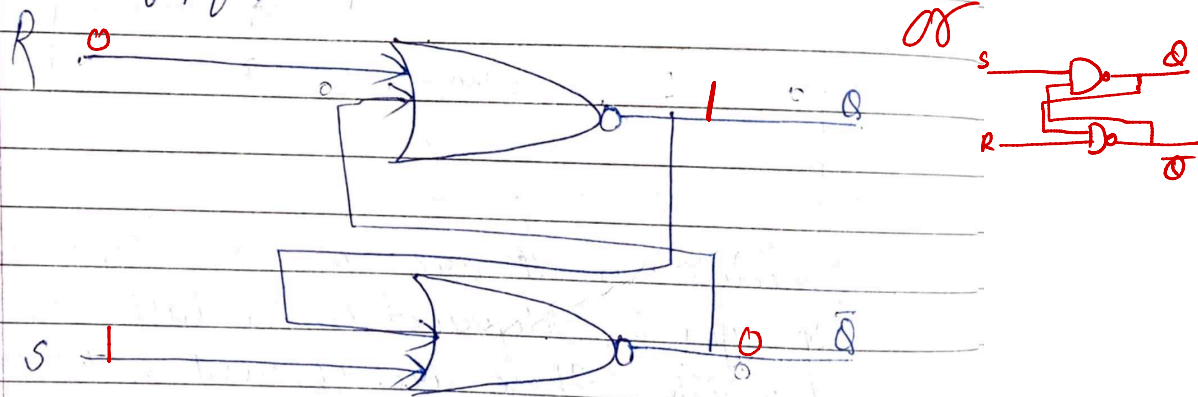
(b) Asynchronous → It is a combinational circuit that has feedback path.

FLIP FLOPS :-

- It is a binary cell which stores one (1) bit of information. It is itself a sequential circuit.
- If the storage element changes its state when clock is exactly one then it is called latch.
- A flip flop changes its state when clock's transition from 0 to 1 (rising edge) or from 1 to 0 (falling edge) and not when clock is one (1).

BASIC FLIP FLOP :-

Basic flip flop is also known as latch. It is designed using two NAND or NOR GATES. It is also known as SR flip flop.



Truth Table (Characteristic Table)

S	R	output
0	0	No change
0	1	Reset / Clear S state
1	0	Set state
1	1	Undefined state.

S	R	Q	Q̄
1	0		

Case 1.

- (a) $Q \rightarrow 0$ (Initially we suppose)
for Input $S=0, R=0$
Result (Q) $\rightarrow 0$

- (b) Initially we suppose $Q \rightarrow 1$
for Input $S=0, R=0$
Result (Q) $\rightarrow 1$

Conclusion:- for Input values $S=0, R=0$, output is same as Q . It means Q will unchanged.

Case 2

- (a) Initially we suppose
 $Q \rightarrow 0$
for Input $S=0, R=1$
Result (Q) $= 0$

- (b) Initially we suppose $Q \rightarrow 1$
for Input $S=0, R=1$
Result (Q) $\rightarrow 0$

Conclusion:- for the Input set $S=0, R=1$, output of the flip flop will be reset or clear state.

Case 3

- (a) Initially we suppose $Q=0$
for Input $S=1, R=0$
Result (Q) $= 1$

- (b) Initially we suppose $Q=1$
for Input $S=1, R=0$
Result (Q) $= 1$

Conclusion:

for the Input Set $S=1, R=0$, Output of the flip flop will be Set state

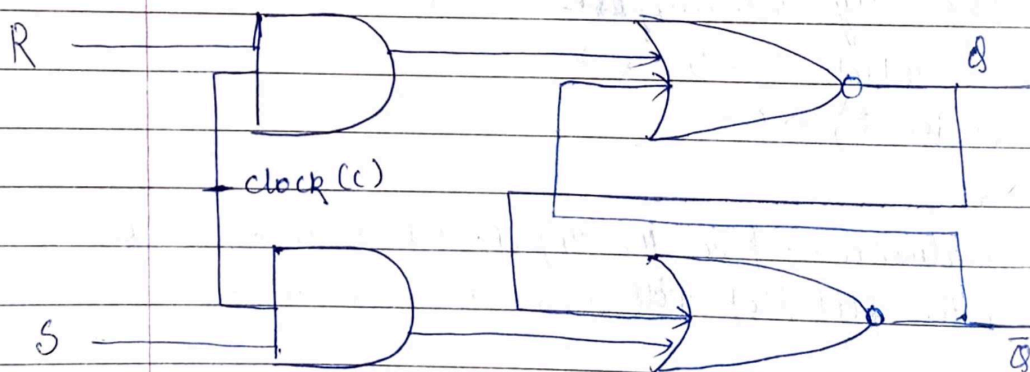
Case 4.

Initially suppose $Q=0, \bar{Q}=1$

for Input Set $S=1, R=1$ output of the flip flop will be undefined state.

RS FLIP FLOP: -

This is the first flip flop that came into existence in the form of flip flop using the basic latch. The main feature of RS flip flop is addition of clock pulse input. In this flip flop changes the state if the clock pulse at that moment is 1



Truth table: -

S	R	output
0	0	No change
0	1	Reset / Clear state
1	0	Set state
1	1	undefined -

Case 1:-(a) for $Q = 0$ Input $S = 0, R = 0$ Result (Q) = 0(b) for $Q = 1$ Input $S = 0, R = 0$ Result (Q) = ~~1~~

Conclusion: The Input $S = 0, R = 0$ then the output of the ^{RS} flip flop will be unchanged.

(b) Case 2:(a) for $Q = 0$ Input $S = 0, R = 1$

Result = 0

(b) for $Q = 1$ Input $S = 0, R = 1$ Result (Q) = 0

Conclusion: The Input $S = 0, R = 1$ then the output will be Reset or clear.

Case 3:(a) for $Q = 0$ Input $S = 1, R = 0$

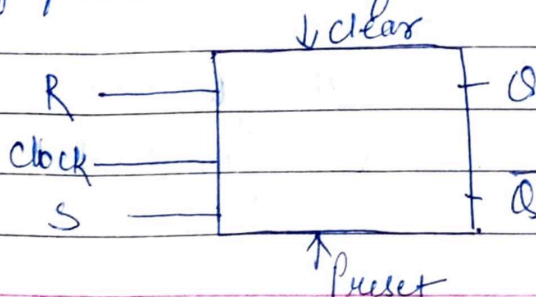
Result = 1

(b) for $Q = 1$ Input $S = 1, R = 0$ Result (Q) = 1

Conclusion: The Input $S = 1, R = 0$, then the output will be Set State.

Case 4

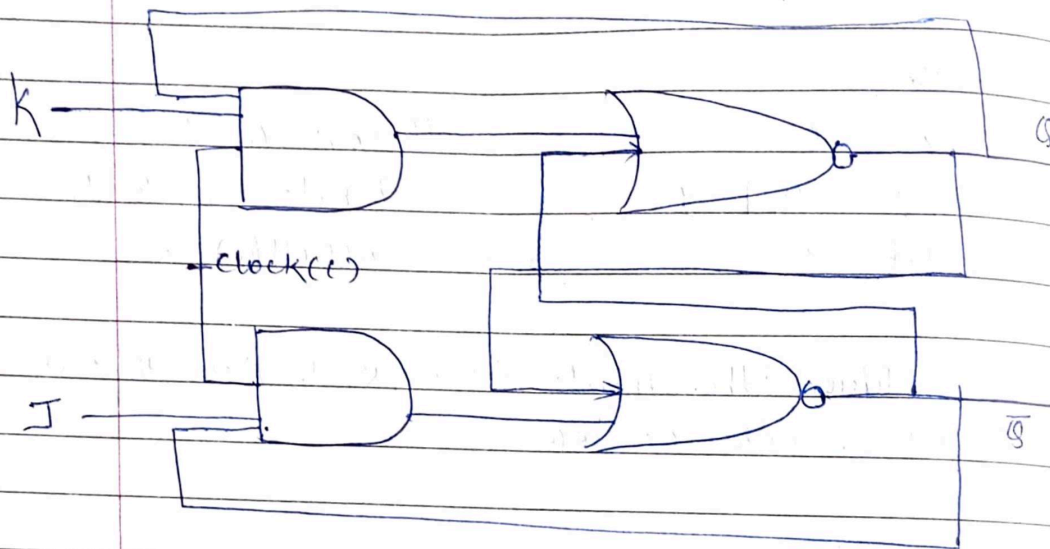
for the Input $R = 1, S = 1$, Output of the flip flop will be undefined.



Block diagram

J-K flip flop :-

The basic drawback with RS flip flop is that one set of input conditions are not utilized. But this can be used with little change in the circuit. In this the last combination is used to complement the state of the flip flop.



Truth Table

J	K	Output
0	0	No change
0	1	Reset/clear
1	0	Set
1	1	Complemented

Case 4.

(a) $Q = 0$

Input $J = 1, K = 1$

Result = 1

(b) $Q = 1$

Input $J = 1, K = 1$

Result = 0

Case 1.

(a) $Q = 0$

for input $j=0, k=0$
Result(Q) = 0

(b) $Q = 1$

for input $j=0, k=0$
Result(Q) = 1

Conclusion :- when the input is $j=0, k=0$ then the output will be unchanged.

Case 2.

(a) $Q = 0$

for input $j=0, k=1$
Result(Q) = 0

(b) $Q = 1$

for input $j=0, k=1$
Result(Q) = 0

Conclusion :- when the input is $j=0, k=1$ then the output will be clear or reset.

Case 3.

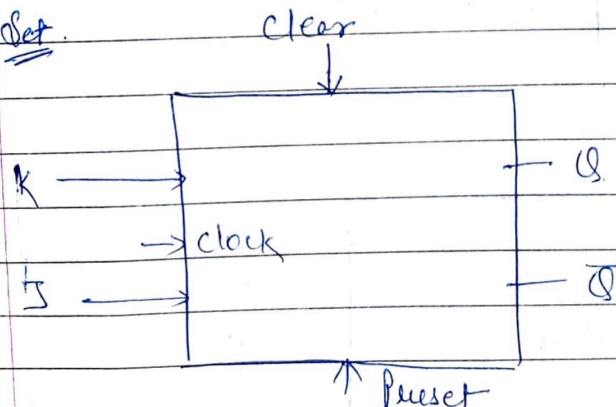
(a) $Q = 0$

Input $j=1, k=0$
Result(Q) = 1

(b) $Q = 1$

Input $j=1, k=0$
Result(Q) = 1.

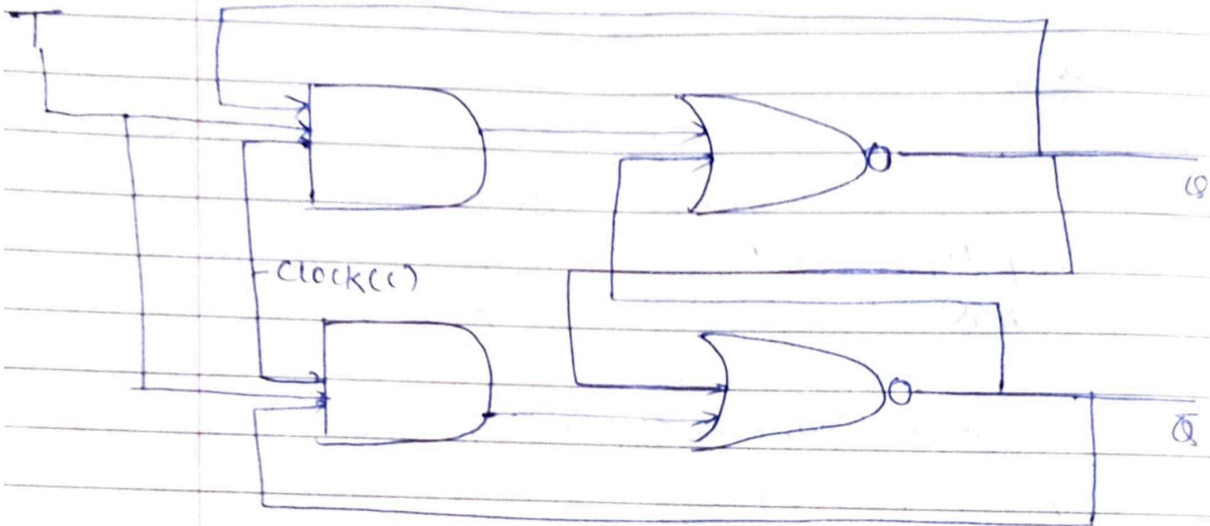
Conclusion Set.



Block diagram for J K flip flop.

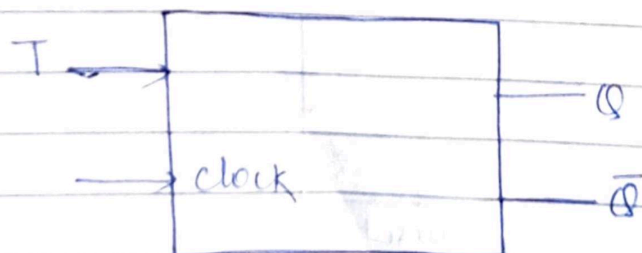
T flip flop :- (Toggle flip flop)

This flip flop is constructed from JK flip flop by joining input J & K together. When $T = 0$, the clock pulse transition does not change the state. When $T = 1$ the clock pulse transition complement the state of flip flop.



Truth table :-

T	output
0	No change
1	Complemented

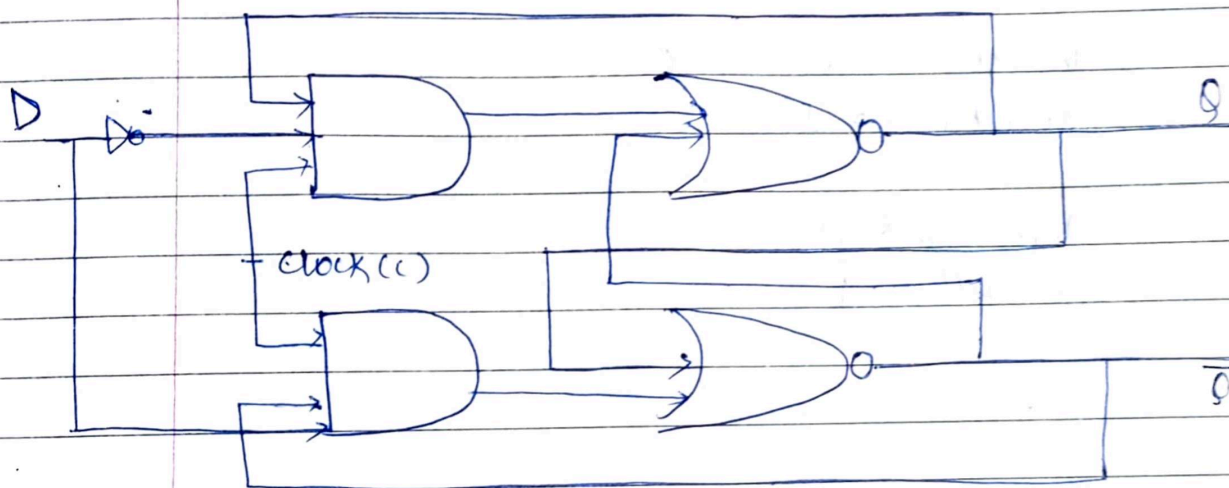


Block diagram for T flip flop

D flip flop (Data or Delay) :-

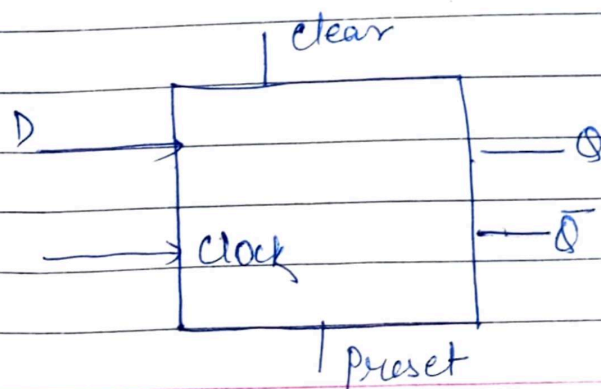
It is a special type of flip flop. The main purpose of this flip flop is to give same output like the same input value. It can store 1 bit of data information so it is known as data flip flop.

It is also referred as delay flip flop because it delays 0 or 1 applied to its input by a single clock pulse.



Truth Table :-

D	Output
0	0
1	1



Block diagram of D flip flop.

Case 1.

$$D = 0$$

(a) Initially suppose $Q = 0$
Result(Q) = 0

(b) Initially Suppose $Q = 1$
Result(Q) = 0

Case 2.

$$D = 1$$

(a) Initially Suppose $Q = 0$
Result(Q) = 1

(b) Initially Suppose $Q = 1$
Result(Q) = 1

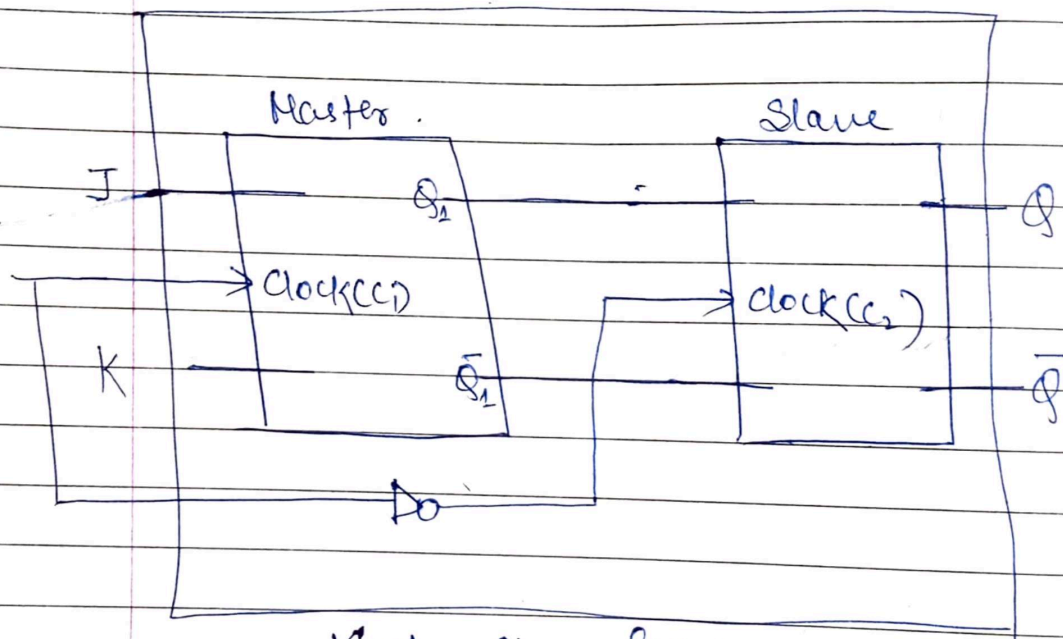
Perform arithmetic addition using signed 2's complement method. assume that maximum size of register is 12 bit.

- (a) -512 and 298 Add
- (b) Subtract 512 & -64
- (c) Add 1025 & 1023

MASTER SLAVE FLIP FLOP:-

It consists of two flip flop. One is the master and other is called slave flip flop. It is the combination of two JK flip flop, in which one flip flop is master and the second flip flop is slave. It is an arrangement of two flip flop connected in such a way that the output of the first flip flop goes into the input of second flip flop.

The clock is set in such a way that the first flip flop gets a direct system clock and second flip flop gets the negation of system clock.



Master Slave flip flop.
Block diagram.

When clock of master $C_1 = 0$ then the clock of slave $C_2(\text{clock}) = 1$.

When clock of master $C_1 = 1$, then the clock of slave $C_2 = 0$.

When a system clock is 1 the first flip flop is enabled and give output Q_1 .

The second flip flop is dependent on the first flip flop. So the second flip flop only have two input.

Truth table for Slave:-

J	K	Output
0	1	clear state
1	0	set state.

Truth table for Master

J	K	output
0	0	No change
0	1	clear
1	0	set
1	1	complemented

Advantage of Master Slave flip flop.

Previous bit holded by master flip flop is contained by the slave next time. We can preserve the value of master flip flop in the slave flip flop.

REGISTERS:-

A register is a group of flip flops that store binary information. In this we also use gates that controls when and how information is transferred to the register. An n bit register has n flipflop and it stores n bit of binary information.